

Mohammed Azif

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Professional Summary

Innovative Robotics Engineer specializing in **ROS 2 Humble**, **Autonomous Navigation (Nav2)**, and software-hardware integration. Expert in developing full-stack robotics applications, including **Flutter-based** control interfaces and **LLM-powered** voice control. Proven experience in high-fidelity simulation and industrial cobot programming with **MoveIt 2**.

Professional Experience

Robotics Engineer

iHub-Data, IIT Hyderabad

Dec 2025 – Present

Hyderabad, India

- **Advanced Navigation Stack:** Enhanced wheelchair autonomy by integrating **2D LiDAR** and **IMU** data to handle complex indoor terrain and wheel-slippage, supported by an **Ultrasonic-based fail-safe layer** for redundant obstacle detection.
- **Navigation Architecture:** Architecting a robust navigation pipeline using **ROS 2 Humble** and **Nav2**, optimized for low-latency performance on **NVIDIA Jetson Orin**.
- **Cross-Platform App Development:** Engineered a custom **Flutter** application for robot teleoperation, featuring secure authentication, real-time telemetry, and destination mapping.
- **AI/LLM Integration:** Implemented **LLM-powered** multi-lingual voice control interfaces to facilitate hands-free navigation commands for users.
- **Digital Twin Simulation:** Developed high-fidelity models in **Gazebo** to benchmark path-planning algorithms against dynamic indoor obstacles before hardware deployment.

Robotics Software Engineer

Alphadroid

Jan 2025 – Dec 2025

Hyderabad, India

- **AMR Fleet Optimization:** Fine-tuned **Nav2 parameters** to eliminate localization drift and improved path smoothness for commercial mobile robot deployments.
- **Cobot Software Stack:** Engineered the software pipeline for a **Doosan M1013** robotic dishwashing system, integrating custom drivers and system logic with **ROS 2 Humble**.
- **Motion Planning:** Leveraged **MoveIt 2** for high-DOF trajectory planning and collision avoidance, achieving a **95% success rate** in pick-and-place trials.
- **Simulation Testing:** Built photorealistic test environments in **NVIDIA Isaac Sim** and **Gazebo** for rigorous SLAM validation and sensor placement analysis.

Technical Skills

Frameworks: ROS 2 (Humble/Foxy), Nav2, MoveIt 2, SLAM, OpenCV, NVIDIA Isaac Sim, Gazebo.

Programming: Python, C++, Dart (Flutter), C, Shell Scripting, SQL.

Hardware Integration: NVIDIA Jetson Orin/Nano, Doosan M1013, LiDAR, IMU, Raspberry Pi.

Tools & OS: Linux (Ubuntu), Docker, Git, URDF/Xacro, Qt, RViz.

Education

B.Tech in Mechatronics Engineering
APJ Abdul Kalam Technological University

May 2024
Kerala, India

- **CGPA:** 7.0/10.0 (First Class Honors)
- **Core Expertise:** Robot Kinematics, Control Systems, Computer Vision, Embedded Systems.

Projects

Smart Wheelchair (Miniature Prototype)

Academic Project

- Developed a ROS-driven prototype utilizing **Raspberry Pi** and **LiDAR** for autonomous indoor mapping and obstacle avoidance with a 7m detection range.